Supplementary Methods and Computational Verification

Submission 6222. Revision prepared 6 October 2026. This supplement records computed results and explicit limits. It contains no invented flow observation or field validation.

S1. Baseline definitions and source integrity

Main coverage baseline: geoBoundaries regional population mask; WorldPop or GHS-POP 2025; EPSG:6933 1-km cells; HOT road snapshot 5 October 2026; main classes; connectors up to 5 km; no deliberate deletion or hub displacement; 100-km coverage radius; fixed 26 existing candidate nodes. The original COD-AB WorldPop count is rechecked separately and does not replace the common-mask optimization results.

population_grid_metadata.json


```
{
  "status": "computed",
  "population_product": "WorldPop 2025 R2025A v1 constrained, alpha",
  "boundary_product": "geoBoundaries COD ADM1 gbOpen boundaryID COD-ADM1-28894549",
  "boundary_note": "alternative to COD-AB, do not interpret as identical administrative mask",
  "provinces": [
    "Kinshasa",
    "Kongo-Central",
    "Kwango",
    "Kwilu",
    "Mai-Ndombe",
    "Mongala",
    "Nord-Ubangi",
    "Sud-Ubangi",
    "Tshuapa",
    "Équateur"
  ],
  "crs": "EPSG:6933",
  "pixel_inclusion": "centre inside union of ten provinces; positive finite pixel counts only",
  "original_selected_pixel_sum": 43302131.740662836,
  "positive_source_pixels": 3651459,
  "positive_cells_1km": 219279,
  "positive_cells_5km": 24553,
  "mass_conservation_relative_error": 1.720603651003945e-16,
  "worldpop_sha256": "141ab3637e858a3597b96b7646dbf6b4c7d258a89a60a5ffcde07f051a109d5e",
  "boundary_sha256": "3deb46256765a4ce78a2e8dadaf6f93f43869877e55d242d1e68139b45a5395a"
}
```


population_product_comparison_metadata.json


```
{
  "worldpop_total": 43302131.74066284,
  "ghspop_total": 34033889.96602864,
  "ghspop_vs_worldpop_pct": -21.403661672228637,
  "positive_ghspop_cells_1km": 147956,
  "positive_ghspop_cells_5km": 21120,
  "cell_5km_pearson": 0.8713868250642843,
  "cell_5km_spearman": 0.9305105678697244,
  "sum_absolute_cell_difference": 23896670.506098725,
  "limits": [
    "both are modelled 2025 population products, not observations",
    "native GHS pixel centres allocated to EPSG6933 bins; grid alignment affects local comparisons",
    "same alternative regional boundary mask with pixel-centre inclusion",
    "GHS SMOD is modelled settlement classification; no functional autonomy inferred"
```

```
    ]
}
```


ROAD_ACCESSIBILITY_METADATA.JSON


```
{
  "snapshot": "2026-10-05",
  "source": "HOT/OSM HDX roads.gpkg",
  "population_mask": "geoBoundaries alternative",
  "lengths": "WGS84 ellipsoidal geodesic, km",
  "junctions": "exact shared source coordinates; interior layer key; endpoints level 0; no spatial snapping between roads",
  "direction": "oneway yes/true/1 forward, -1 reverse; all other values assumed bidirectional, including missing/reversible",
  "limits": [
    "no turn restrictions or access tags",
    "vertex attachment rather than nearest segment",
    "off-network straight connectors are hypothetical, not observed traversability",
    "missing layer/bridge tags may cause incorrect junctions",
    "roads selected by ten provincial pcodes; outside-province detours omitted",
    "no rail/fluvial routing or navigability validation",
    "26 hubs are existing urban nodes; this is not a new-hub optimization"
  ],
  "scenarios": [
    {
      "scenario": "road_classes",
      "classes": [
        "living_street",
        "primary",
        "primary_link",
        "residential",
        "road",
        "secondary",
        "secondary_link",
        "service",
        "tertiary",
        "tertiary_link",
        "trunk",
        "trunk_link",
        "unclassified"
      ],
      "oneway_nonempty_features": 1589,
      "oneway_yes_features": 1378,
      "layer_nonempty_features": 1353,
      "bridge_nonempty_features": 1332,
      "weak_component_sizes_top10": [
        1294595,
        187112,
        47697,
        37938,
        21865,
        18329,
        15393,
        12234,
        8893,
        4102
      ]
    },
    {
      "scenario": "including_tracks",
      "classes": [
        "living_street",
        "primary",
```

```
                "primary_link",
                "residential",
                "road",
                "secondary",
                "secondary_link",
                "service",
                "tertiary",
                "tertiary_link",
                "track",
                "trunk",
                "trunk_link",
                "unclassified"
            ],
            "oneway_nonempty_features": 1592,
            "oneway_yes_features": 1380,
            "layer_nonempty_features": 1420,
            "bridge_nonempty_features": 1398,
            "weak_component_sizes_top10": [
                1844769,
                300759,
                57381,
                45489,
                23308,
                20042,
                19192,
                16667,
                15659,
                5790
            ]
        }
    ]
}
```


JOINT_COVERAGE_METADATA.JSON


```
{
    "completed_scenarios_per_p": 96,
    "total_optimization_runs": 288,
    "objective": "maximize modelled population within 100 km road+assumed connector distance",
    "candidate_set": "26 existing GeoNames urban nodes, not proposed new cities",
    "factors": {
        "population": [
            "WorldPop2025",
            "GHSPOP2025"
        ],
        "grid_km": [
            1,
            5
        ],
        "tracks": [
            false,
            true
        ],
        "connector_km": [
            1,
            2.5,
            5
        ],
        "road_feature_removal_fraction": [
            0,
            0.01
```

```
    ],
    "hub_coordinate_displacement_km": [
      0,
      0.25
    ]
  },
  "perturbation_seed": 20261006,
  "removal_method": "one independent uniform draw per selected road feature; fixed sample for each graph scenario; not calibrated OSM incompleteness",
  "coordinate_method": "fixed 250 m displacement with independent random azimuth per hub; stress test not positional error distribution",
  "limits": [
    "scenario ranges are not confidence intervals",
    "objective differs from legacy p-median",
    "alternative population masks and products are modelled",
    "equal-objective site sets may be nonunique; reported identities are one optimum",
    "not a Sobol decomposition",
    "no travel speeds because no travel-time claims",
    "no observed-flow validation"
  ]
}
```


EXACT_PMEDIAN_METADATA.JSON


```
{
  "demand_total": 7371525.750308432,
  "demand_definition": "sum of released 5-km node-buffer population estimates, not the regional population",
  "nodes": 26,
  "edges": 49,
  "distance": "released rounded edge distances in km",
  "solver": "scipy.optimize.milp / HiGHS"
}
```


CANDIDATE_WEIGHT_AUDIT_METADATA.JSON


```
{
  "candidate_count": 75,
  "models": [
    "reference seven criteria",
    "uniform seven criteria"
  ],
  "dirichlet_concentration": 60,
  "simulations_per_model": 5000,
  "seed": 20260902,
  "limits": [
    "only 75 previously sampled and filtered sites, not all eligible regional land",
    "one winner per province; no across-province separation constraint",
    "weights are assumed, not elicited expert judgments",
    "weight-only frequencies are conditional, not probabilities of feasibility",
    "protected areas, tenure and infrastructure capacity unverified"
  ]
}
```


S2. BOUNDARY COUNT COMPARISON

COD-AB WorldPop selected count = 43,227,817.79641058. Common geoBoundaries count = 43,302,131.74066284. Difference = 74,313.94425226 (0.17191878% of COD-AB total). This comparison verifies the earlier total and is not a second regional optimization under COD-AB.

S3. All six road-reachability configurations

{"scenario": "road_classes", "max_connector_km": 1.0, "road_features": 106892, "graph_vertices": 1734033, "directed_arcs": 3594763, "weak_components": 1273, "eligible_hubs": 25, "population_total": 43302131.74066284, "population_attached_pct": 88.31512969232324, "population_reachable_pct": 78.4931613457577, "population_attached_no_hub_path_pct": 9.821968346565535, "population_unattached_pct": 11.68487030767675, "reachable_population_mean_distance_km": 76.58858441574219, "population_within100km_pct": 55.58750360632013}

{"scenario": "road_classes", "max_connector_km": 2.5, "road_features": 106892, "graph_vertices": 1734033, "directed_arcs": 3594763, "weak_components": 1273, "eligible_hubs": 26, "population_total": 43302131.74066284, "population_attached_pct": 92.51683193091556, "population_reachable_pct": 82.02619927898995, "population_attached_no_hub_path_pct": 10.4906326519256, "population_unattached_pct": 7.483168069084438, "reachable_population_mean_distance_km": 77.6537383250754, "population_within100km_pct": 58.10959650715915}

{"scenario": "road_classes", "max_connector_km": 5.0, "road_features": 106892, "graph_vertices": 1734033, "directed_arcs": 3594763, "weak_components": 1273, "eligible_hubs": 26, "population_total": 43302131.74066284, "population_attached_pct": 96.13577926081923, "population_reachable_pct": 84.95129616127578, "population_attached_no_hub_path_pct": 11.184483099543453, "population_unattached_pct": 3.864220739180741, "reachable_population_mean_distance_km": 79.820525012691, "population_within100km_pct": 59.37470564073577}

{"scenario": "including_tracks", "max_connector_km": 1.0, "road_features": 147365, "graph_vertices": 2518218, "directed_arcs": 5187173, "weak_components": 2779, "eligible_hubs": 26, "population_total": 43302131.74066284, "population_attached_pct": 91.35801905655045, "population_reachable_pct": 81.43547394073144, "population_attached_no_hub_path_pct": 9.922545115819007, "population_unattached_pct": 8.641980943449537, "reachable_population_mean_distance_km": 76.02766316124873, "population_within100km_pct": 57.59358852077848}

{"scenario": "including_tracks", "max_connector_km": 2.5, "road_features": 147365, "graph_vertices": 2518218, "directed_arcs": 5187173, "weak_components": 2779, "eligible_hubs": 26, "population_total": 43302131.74066284, "population_attached_pct": 94.61709499687294, "population_reachable_pct": 84.16367984061533, "population_attached_no_hub_path_pct": 10.453415156257618, "population_unattached_pct": 5.382905003127044, "reachable_population_mean_distance_km": 77.43782279262729, "population_within100km_pct": 59.07054405524485}

{"scenario": "including_tracks", "max_connector_km": 5.0, "road_features": 147365, "graph_vertices": 2518218, "directed_arcs": 5187173, "weak_components": 2779, "eligible_hubs": 26, "population_total": 43302131.74066284, "population_attached_pct": 97.18304446656111, "population_reachable_pct": 86.1637486531681, "population_attached_no_hub_path_pct": 11.019295813393022, "population_unattached_pct": 2.816955533438859, "reachable_population_mean_distance_km": 79.12641692639068, "population_within100km_pct": 59.89509090659399}

S4. Complete audit tables and optimization runs

The replication package provides machine-readable topology metrics for all nodes and all graph variants; exact p-median outcomes for p = 2–20; 288 joint maximal-covering result rows and selected-set rows; all candidate weight-centre comparisons; settlement classifications; population-product comparisons; and source metadata. These files are the numerical evidence underlying the manuscript tables. The plot code in build_final.py reads the same tables. A zero solver gap establishes an optimum for the encoded candidate set, not unique facilities or empirical correctness.

S5. Secondary observational source assessed, not used as validation

JICA, PDTK Final Report, April 2019, vol. 1, part 1: section 2.2.7, printed pp. 2-47–2-48, discusses GPS travel-speed surveys in 2017. Table 2.2.16 lists 13 Kinshasa urban routes. The source PDF SHA-256 is b0f5aa1a9b3f4845daa03afbbf8fe6347ade8ecf6e68902148bc38ea44108599. The retained route inventory is descriptive. No route matching, travel-speed digitization, or regional predictive validation has been performed. The report does not supply an OD target for the ten-province maximal-covering problem.

S6. Reproduction sequence and interpretation

Install the Python dependencies recorded in requirements_revision.txt. Obtain the original released inputs and download the raw files to revision_sources using the exact URLs in source_manifest.json. Preserve the raw snapshots and compare their checksums before recomputation. Run population aggregation, GHS comparison, topology audit, exact p-median audit, candidate-weight audit, road-accessibility processing, joint coverage, and the boundary check in the order specified in README_REVISION.md. The road graph has millions of vertices; sufficient memory and disk space are required. Download failure or different producer content must be reported, not silently replaced with simulated data.

The generated source counts and optimization outputs were checked in the present environment. A second independent clean-machine reproduction and matched empirical mobility validation have not been completed. The package supplies the materials for that reproduction rather than certifying an unperformed independent run.